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Trans-Fast MHD Winds in a Pulsar Magnetosphere

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Abstract

We consider the problem of how a pulsar wind dominated by electro-magnetic energy converts itself into a kinetic energy-dominated wind, that is required by magnetohydrodynamical (MHD) models of interaction between the pulsar and the nebula. It is known that stationary, ideal-MHD wind models based on radial configurations are incapable of explaining the energy conversion from the electro-magnetic energy to the plasma's kinetic energy. For *non-radial* field geometries, we consider a detailed analysis of the critical conditions of trans-fast MHD winds and discuss the effects of the magnetic field on the wind acceleration. Although the flow properties associate with the magnetic flux tube divergence, we survey the acceptable ranges of the wind parameters as a pulsar wind. Further, we demonstrate the Crab pulsar wind with a magnetic flux tube diverging as $\propto x^{-0.4}$. We see that the rapid acceleration is caused by a nearby Alfvén point; the velocity is almost terminated within several thouthand times of the fast magnetosonic point; thus, the kinetic energy-dominated wind is realized.

Key words: Magnetohydrodynamics — Nebulae: individual (Crab Nebula) — Pulsars — Stars: winds

1. Introduction

The luminosity of the Crab Nebula is compatible to the rotational energy loss rate estimated from the observed spin-down rate of the central pulsar. It is clear that the power source for the Crab Nebula is the pulsar located within. The pulsar's energy output is converted into the nebular synchrotron luminosity with high efficiently, 10%–20%. The pulsar also provides the nebula with its strong troidal magnetic field (e.g., see Kennel, Coroniti 1984a). The pulsar and nebula therefore form a tightly coupled hydromagnetic system. In constructing a model of the pulsar–nebular interaction, Rees and Gunn (1974) assumed that the pulsar's spin-down luminosity is transported by a mixed relativistic wind-electromagnetic wave with an unspecified ratio between the two energy fluxes (see also Kundt, Krotscheck 1980). The development of a theory of pair production in pulsar magnetospheres suggests that the pair density necessarily exceeds the Goldreich–Julian (1969) charge density, which neutralizes the parallel component of the pulsar's vacuum electric field. This fact suspects that the strong pulsar wave cannot propagate in a dense plasma (Max, Perkins 1971; Kennel et al. 1973; Kennel, Pellat 1976). We are thus led to reconsider the concept of Rees–Gunn's pulsar–nebular interaction model.

We discuss the structure of a pulsar magnetosphere enclosed by the nebula. This magnetosphere is divided into three regions for the Crab pulsar–nebular system: (i) the plasma source located near to the star surface ($r_{\rm i}^{\rm Crab} \approx 10^4$ m), where electron–positron pairs are created, (ii) the wind region that is outside the plasma source and extends beyond the light cylinder ($r > r_{\rm L}^{\rm Crab} \approx 1.5 \times 10^6$ m for a spin frequency of $\Omega_F^{\rm Crab} \approx 200$ rad s^{-1}), where a highly relativistic wind blows away toward the nebula, and is terminated by an magnetohydrodynamical (MHD) reverse shock at $r_{\rm S}^{\rm Crab} \approx 3 \times 10^{15}$m $= 0.1$ pc, and (iii) the nebula ($r_{\rm S}^{\rm Crab} < r < r_{\rm N}^{\rm Crab} \approx 2$ pc). Regions (i) and (iii) present the inner and outer boundary conditions of the wind, while coupling the theoretical and observational facts. We then discuss the pulsar–nebular interaction by the MHD wind.

The Crab pulsar's magnetosphere should be energetically and dynamically dominated by the strong neutron star's magnetic field, at least out to the light cylinder. There are strong reasons to believe that pulsars create an electron–positron pair plasma on open current-carrying magnetic field lines that connect the polar cap of a magnetized neutron star to its exterior magnetosphere (Sturrock 1971; Ruderman, Sutherland 1975; Arons 1979). Unfortunately, where in the magnetosphere are pairs created, and how many are created, are

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questions still not settled. The escaping pair density depends on the number of generation of pairs that are photoproduced, and differs from model to model. However, it seems clear that the charge of plasma is neutral once it leaves the charge-separated region of pair production near to the pulsar. That is, the pair density necessarily exceeds the Goldreich–Julian charge density. Thus, the multiplicity of an electron–positron plasma is estimated at $\sim 10^3$–10^5 (Tademaru 1973; Ruderman, Sutherland 1975; Kundt 1981; Daugherty, Harding 1982; Gurevich, Istomin 1985; Cheng et al. 1986; Mestel, Shibata 1994).

The ratio of the Poynting flux to the particle kinetic-energy flux of the wind is expected to be $\sigma_i \sim 10^4$–10^5 while leaving the pair-creating region. The escaping electron–positron outflow has a initial poloidal four-velocity u_i of $\sim 10^2$–10^3 (Mestel, Shibata 1994; see also Michel 1982 and references therein). This outflow should be accelerated in the pulsar wind region (ii) to form the nebula's strong MHD shock, where the flow is decelerated and the gas pressure of the flow increases to match the boundary conditions at the nebula. It is necessary to include the weak magnetization of the wind to satisfy the boundary conditions and to calculate the nebular synchrotron radiation self-consistently; then, the asymptotic poloidal velocity u_S is estimated from a synchrotron model of the nebula's optical to γ-ray emission, which yield the wind's upstream Lorentz factor of $\gamma_S \approx 3 \times 10^6$ and the very small ratio of $\sigma_S \approx 0.0033$ (Kennel, Coroniti 1984a,b).

For a simple spherical symmetric wind, Kennel, Fujimura, and Okamoto (1983) surveyed the dependence of a finite-temperature adiabatic MHD wind on a wide range of source parameters, so as to enable wind theory to respond flexibly to the variety of pulsar source models that had been proposed. They found that for a radial configuration of magnetic field lines, the fast Mach-number of cold winds is at most unity, while hot winds can achieve a large fast Mach-number. In the case of a rapidly rotating pulsar magnetosphere, however, a hot plasma would quickly cool down, and a strong thermal wind would not flow from the rapidly rotating pulsar, because of strong synchrotron cooling (e.g., Shibata 1991). Thus, since the radial configuration cannot explain the observed wind acceleration, a strong reverse shock cannot stand.

Michel (1969) treated a zero-temperature wind in a radial configuration and Kennel et al. (1983) showed that the most accelerated wind solution (the Michel's 'minimum energy' solution) reaches the fast magnetosonic point located at infinity with a terminal wind velocity of $u_\infty \sim (E/\mu)^{1/3}$, which means that only a small part of the energy is in the plasma, and that the Poynting flux dominates the total energy E. For the wind acceleration, however, the geometry of magnetic field line is essential. For example, Takahashi (1993) showed that, under a magnetic field line which slightly bends toward the rotation axis, a trans-fast MHD wind is available, where the X-type fast magnetosonic point appears at a finite distance; Begelman and Li (1994) argued that if the flux tube in a wind diverges even slightly faster than that in a radial wind, the X-type fast magnetosonic point appears at a finite distance, and because of the "magnetic nozzle" effect the wind can accelerate, converting the Poynting flux to the kinetic-energy flux.

Because the wind properties depend on the magnetic field configuration, we need to discuss the pulsar wind based on the magnetic field geometry. On the other hand, the magnetic field configuration is affected by the inertia of the streaming wind. Thus, we should solve the pulsar magnetosphere self-consistently by the poloidal equation and the force-balance equation under the boundary conditions at the plasma source. The structure of the pulsar magnetosphere has been developed by many authors (e.g., Sulkanen, Lovelace 1990; Beskin et al. 1993; Mestel, Shibata 1994). Unfortunately, we have not yet obtained a self-consistent model of the magnetosphere, because we do not have sufficient information about the boundary conditions of the magnetosphere. Hereafter, we only consider the MHD wind along *given* magnetic field lines. Of course, in general, this may be inconsistent with the force-balance equation. However, we believe that we will clarify magnetic field contributions on the MHD winds.

In section 2, we begin by considering relativistic ideal-MHD winds in a stationary and axisymmetric rapidly rotating magnetosphere, and present the critical conditions of a trans-fast MHD wind in the zero-temperature (cold) limit. The trans-fast MHD wind is specified by the location of the Alfvén point, the light cylinder, and the fast magnetosonic point, instead of the four field-aligned constant quantities of the wind: the total energy and angular momentum of the wind, the angular frequency of magnetic field line, and the particle number flux per magnetic flux.

In section 3, to clarify the effects of the magnetic field geometry on the wind acceleration, we survey the dependence of relativistic trans-fast MHD winds on a wide range of source parameters. Then, we see that to realize efficient acceleration of the trans-fast MHD wind there must be a region where $(x^2 B_p)' < 0$, *at least between the Alfvén point and the fast magnetosonic point*, where B_p is the poloidal magnetic field, x is the radius from the rotation axis, and the prime denotes the derivative along a stream line.

In section 4, under *given* configuration of magnetic field lines, $B_p \propto x^{-(2+\delta)}$ with constant δ, we try to demonstrate the highly-accelerated pulsar wind, and investigate whether a magnetically dominated (large-σ) wind can convert into a particle kinetic energy-dominated (low-σ) wind between the pulsar and the nebular shock. For the Crab pulsar's wind parameters, we obtain a

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pulsar wind solution accelerated from $\gamma_{\rm i} \sim 10^2$ to $\gamma_{\rm S} \sim 10^6$, and at the same time the magnetization parameter decreases from $\sigma_{\rm i} \sim 10^4$ to $\sigma_{\rm S} \sim 10^{-3}$. Significant acceleration starts from the Alfvén point and continues far beyond the fast magnetosonic point; however, the velocity of the wind reaches the terminal velocity far inside of the nebular shock. As mentioned by Begelman and Li (1994) to explain the energy transport from the Poynting flux to the kinetic energy flux, the divergence of a flux tube $x^2 B_p$ must have a dependence of $\sim x^{-\delta}$ with a positive $\delta > 0$, at least far outside of the fast magnetosonic point. In fact, the energy transport occurs beyond the Alfvén point; the equipartition of energy is achieved at several times the fast magnetosonic point; beyond this radius σ decreases as $\sigma \propto x^{-0.4}$ for the Crab pulsar wind. This demonstration is consistent with both the pair-creation model and the synchrotron nebula model.

Finally, in section 5 we discuss the magnetic field structure of a magnetosphere to generate the kinetic-energy dominated trans-fast MHD wind.

2. Critical Conditions for Trans-Fast MHD Winds

In this section we present the basic equations of a stationary and axisymmetric ideal-MHD wind and derive the conditions for a trans-fast MHD wind in the cold limit. Here, we can ignore the gravitational field produced by the central star and use the cylindrical coordinates (t, R, ϕ, Z) with $c = 1$ unit.

The structure of the magnetosphere is determined by both the so-called Bernoulli equation along magnetic field lines and the force-balance equation across the magnetic field lines. These equations are derived from the equation of motion for a plasma,

$$T^{\mu\nu}{}_{,\nu} = \left[(\rho + P)u^\mu u^\nu - P\eta^{\mu\nu} + T^{\mu\nu}_{\rm EM}\right]_{,\nu} = 0\,, \tag{1}$$

the conservation law of particle number $(nu^\mu)_{,\mu} = 0$, the ideal-MHD condition $u^\mu F_{\mu\nu} = 0$, and Maxwell's equations. Here, ρ, P, and n are the total energy density, the pressure of the plasma, and the proper particle density, respectively; $T^{\mu\nu}_{\rm EM}$ is the electromagnetic part of the energy momentum tensor; $F_{\mu\nu}$ is the electromagnetic field tensor; u^μ is the four-velocity of the plasma.

Stationariness and axisymmetry of the ideal MHD flows provide the existence of a stream function, $\psi = \psi(R, Z)$; both the poloidal flow and the magnetic field follow curves with $\psi =$ constant. There are four field-aligned constants, which are the total energy $E(\psi)$, the total angular momentum $L(\psi)$, the angular velocity of the field line $\Omega_F(\psi)$, and the particle flux in unit magnetic flux $\eta(\psi)$. These conserved quantities are defined by (e.g., Bekenstein, Oron 1978; Camenzind 1986a)

$$E = \mu u_t - \frac{\Omega_F B_\phi}{4\pi\eta}, \tag{2}$$

$$L = -\mu u_\phi - \frac{B_\phi}{4\pi\eta}, \tag{3}$$

$$\Omega_F = \frac{F_{tR}}{F_{R\phi}} = \frac{F_{tZ}}{F_{Z\phi}}, \tag{4}$$

$$\eta = \frac{nu_p}{B_p} = \frac{nu^t}{B^\phi}(\Omega - \Omega_F), \tag{5}$$

where $u_p \equiv [(u^R)^2 + (u^Z)^2]^{1/2}$ is the poloidal velocity; $B_p \equiv [(B^R)^2 + (B^Z)^2]^{1/2}$ is the poloidal magnetic field; B^R and B^Z are the component of the magnetic field seen by a lab-frame observer, which are written by $B^R \equiv -(\partial\psi/\partial Z)/R$ and $B^Z \equiv (\partial\psi/\partial R)/R$; and $B_\phi \equiv RF_{RZ}$ is the toroidal magnetic field; $\Omega \equiv u^\phi/u^t$ is the angular velocity of the plasma and μ is the specific enthalpy (in the cold MHD limit $\mu = m_{\rm p}$, where $m_{\rm p}$ is the particle's mass). It is convenient to normalize the distances in unit of the light cylinder radius $R_{\rm L} = 1/\Omega_F$; that is, $x = R/R_{\rm L}$ and $z = Z/R_{\rm L}$, then the radius of $x = 1$ denotes the light cylinder.

The poloidal wind equation is derived from the normalized condition $u_\alpha u^\alpha = 1$ in terms of the four conserved quantities (e.g., Camenzind 1986b; Takahashi 1993):

$$\mu^2(1 + u_p^2) = E^2\left\{(1 - x^2 - 2M^2)\mathcal{A}^2 + \left[x^2 - (\Omega_F L/E)^2\right]\right\}/x^2, \tag{6}$$

where the function $\mathcal{A}$ is introduced as

$$\mathcal{A} \equiv \frac{x^2 - (\Omega_F L/E)}{M^2 - 1 + x^2} \tag{7}$$

and the relativistic Alfvén Mach-number M is defined by

$$M^2 \equiv \frac{4\pi\mu n u_p^2}{B_p^2} = \frac{4\pi\mu\eta}{\Phi} x^2 u_p, \tag{8}$$

with $\Phi(x;\psi) \equiv x^2 B_p = x|\nabla\psi|$, which denotes the divergence of a magnetic flux tube. The function $\mathcal{A}$ in equation (6) seems to be singular at the Alfvén point $x_{\rm A}$, where $M_{\rm A}^2 = 1 - x_{\rm A}^2$; here, the poloidal velocity u_p is equal to the relativistic Alfvén wave speed $u_{\rm AW}$ (see Camenzind 1986b). To reach the far distant nebula, however, an ejected wind from a plasma source must pass through the Alfvén point $(x_{\rm A}, u_{\rm A})$ smoothly on the (x, u_p)-plane, where $u_{\rm A} \equiv u_p(x_{\rm A}) = u_{\rm AW}(x_{\rm A})$. Thus, the physical wind solution must satisfy the condition $\Omega_F L/E = x_{\rm A}^2$ at the Alfvén point. Then, the function $\mathcal{A}$ is finite at the Alfvén point and always has a positive value in whole x. (We should note that the Alfvén point is the focus of a bundle of solutions, and plays no role in selecting a particular solution; thus, we cannot specify a wind solution.)

To plot a poloidal wind solution on the (x, u_p)-plane, we reduce equation (6) to

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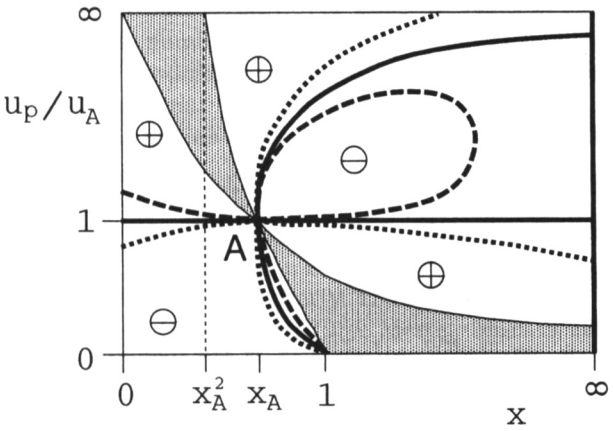


Fig. 1. Schematic picture of the three solutions $M^2 = M^2_{N1}$, $M^2 = M^2_{N2}$, and $M^2 = M^2_{N3}$ on the (x, u_p)-diagram. The thick curves denote the solutions for the radial configuration, which satisfy $(\ln \Phi)' = 0$ and are obtained from $P_1 = 0$ or $P_2 = 0$. The dotted curves and broken curves denote the solutions for $(\ln \Phi)' > 0$ and $(\ln \Phi)' < 0$, respectively. Here, the values of L/E and Ω_F are fixed for each configuration, so that the Alfvén radius and the light cylinder are located at a same place on the (x, u_p)-plane, respectivery; and the scale of vertical axis is normalized by u_A. The symbols of $\oplus$ and $\ominus$ are positive N and negative N regions, respectively. In the case of $(\ln \Phi)' < 0$, the second term in square bracket of equation (12) becomes positive, so the $N = 0$ lines shift into the $\ominus$-region as compared with the $(\ln \Phi)' = 0$ case. Then, we see that the lines of $N = 0$ make a loop shrink with the regression of $(\ln \Phi)'$, while for $(\ln \Phi)' > 0$ the area between $M^2 = M^2_{N2}$ and $M^2 = M^2_{N3}$ expands into the $\oplus$-region with the progression of $(\ln \Phi)'$. The shaded regions are forbidden regions, where no wind solutions exist.

$$\left(\frac{E}{\mu}\right)^2 = \frac{x^2(1+u_p^2)(M^2-1+x^2)^2}{(x^2-x_A^4)(M^2-M^2_{[-]})(M^2-M^2_{[+]})}, \tag{9}$$

where

$$M^2_{[\pm]} = \frac{x^2(1-x_A^2)^2 \pm x(1-x_A^2)|x^2-x_A^2|}{x^2-x_A^4}. \tag{10}$$

When x_A, η, and Φ are specified, the poloidal velocity $u_p(x)$ is represented as a level contour of $E(x, u_p)$, which is plotted by using equation (9) with relation (8). We should note that, as shown in figure 1 by the shaded regions on the (x, u_p)-plane, there is a pair of wedge-shaped 'forbidden regions,' where $E^2 < 0$, and, therefore, we cannot give any timelike flow; the forbidden regions are $M^2_{[-]} \le M^2 \le M^2_{[+]}$ in $x_A^2 < x \le 1$; $M^2_{[-]} \le M^2$ in $x \le x_A^2$; and $0 \le M^2 \le M^2_{[+]}$ in $x > 1$. We next consider the curve $M^2 = 1 - x^2$, where E^2 vanishes, except for at the Alfvén point, on the (x, u_p)-plane. Since this curve gives $u_p = u_{AW}$, we can call the region $M^2 < 1 - x^2$ the sub-Alfvénic region and call the region $M^2 > 1 - x^2$ the super-Alfvénic region. Because the inequalities $M^2_{[-]} < 1 - x^2 < M^2_{[+]}$ in $x > x_A$ and $M^2_{[-]} < 1 - x^2$ in $x \le x_A$ are satisfied, we can say that the forbidden regions separate the sub-Alfvénic region from the super-Alfvénic region in the (x, u_p)-plane. At the Alfvén point, the equalities $M^2_{[-]A} = 1 - x_A^2 = M^2_{[+]A}$ keep the flow's energy finite.

After passing through the Alfvén point, the MHD wind would pass through the fast magnetosonic point, and would then go to the nebula. We now discuss the critical conditions for the trans-fast MHD wind. By differentiating equation (6) along a streamline, the poloidal differential wind equation is written by

$$(\ln u_p)' = \frac{N}{D_{AW} D_{FM}}, \tag{11}$$

where the prime denotes the derivative along a streamline, $\partial_x + (B^z/B^x)\partial_z$, and

$$N = \left(\frac{E}{\mu}\right)^2 \left[P_1 P_2 - M^4(x^2-x_A^2)^2 x(\ln \Phi)'\right] \frac{1}{x^3}, \tag{12}$$

$$D_{AW} = (M^2 - 1 + x^2)^2, \tag{13}$$

$$D_{FM} = \left(\frac{E}{\mu}\right)^2 \left(\frac{\mathcal{A}}{x}\right)^2 M^4 - u_p^2(M^2 - 1 + x^2), \tag{14}$$

with

$$P_1 \equiv x_A^2(M^2 - 1 + x^2) - (x^2 - x_A^2), \tag{15}$$

$$P_2 \equiv (M^2 - 1 + x^2)P_1 - 2M^2(x^2 - x_A^2). \tag{16}$$

The differential wind equation (11) seems to become singular at the Alfvén point, where $D_{AW} = 0$, and the fast magnetosonic point, where $D_{FM} = 0$; at these points, the poloidal velocity equals to the relativistic Alfvén wave speed, and the relativistic fast magnetosonic wave speed, respectively. In the cold MHD wind limit, since the slow magnetosonic wave speed is zero, we do not discuss the critical condition at the slow magnetosonic point.

To realize a physical trans-fast MHD wind, we must require $D_{FM} = N = 0$ at the fast magnetosonic point. From the condition $D_{FM} = 0$, the energy flux of a trans-fast MHD wind is given by

$$(\eta E)_* = \left[\frac{B_p(M^2 - 1 + x^2)^{1/2}}{4\pi(\mathcal{A}/x)}\right]_F, \tag{17}$$

where the subscript 'F' denotes the quantities at the fast magnetosonic point and the subscript '$*$' denotes the quantities of a trans-fast MHD flow. The Alfvén Mach-number at the fast magnetosonic point, denoted by $M_F^2 \equiv M^2(x_F; x_A, \Phi)$, is the solution for $N = 0$. Combining the poloidal equation (6) and equation (17), we also obtain the total energy E_* of the trans-fast MHD wind,

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$$\left(\frac{E_*}{\mu}\right) = \left[\frac{(M^2-1+x^2)^{1/2}}{(\mathcal{A}/x)Q^{1/2}}\right]_{\rm F}, \tag{18}$$

and the particle flux through a flux-tube η_*,

$$\eta_* = \left[\frac{B_p}{4\pi\mu}Q^{1/2}\right]_{\rm F}, \tag{19}$$

where

$$Q \equiv (M^2-1+x^2)^2\frac{M^2-M_{\rm A}^2}{x^2-x_{\rm A}^2}+M^2(1-x^2-2M^2)\,. \tag{20}$$

Thus, we can express the conserved quantities, η_* and E_*, of the trans-fast MHD wind in terms of $x_{\rm A}$, $x_{\rm F}$, and $\Phi(x;\psi)$. We should note that the behavior of $M_{\rm F}^2$ is essential to determine the existence and nature of the trans-fast MHD wind solution, which strongly depends on the magnetic field configuration. In the next section we consider the effects of the magnetic field configuration on the behavior of $M_{\rm F}^2$.

3. Effects of Field Geometry on Fast Magnetosonic Point

In this section we search for acceptable ranges of the wind parameters for the trans-fast MHD wind. First, we consider the behavior of $M_{\rm F}^2(x_{\rm F};x_{\rm A},\Phi)$, which is the solution of $N(x,M^2;x_{\rm A},\Phi)=0$. In the cold limit, the equation $N=0$ is a cubic equation of M^2 under the given $x_{\rm A}$ and $\Phi(x;\psi)$. The discriminant $\mathcal{D}_N(x;x_{\rm A},\Phi)$ of $N(x,M^2;x_{\rm A},\Phi)=0$ is

$$\begin{aligned}\mathcal{D}_N = -\frac{x^4\alpha^2\varepsilon^3}{27x_{\rm A}^{16}}\Bigg[&\left(\varepsilon+8x^2\alpha\right)x_{\rm A}^4\\ &+3\varepsilon\left(\varepsilon+5x^2\alpha\right)x_{\rm A}^2\,x(\ln\Phi)'\\ &+\frac{3}{4}\varepsilon\left\{[\varepsilon+x_{\rm A}^2(x^2-1)^2]^2+2\alpha^2x^4\right\},[x(\ln\Phi)']^2\\ &+(x^2-1)^2\varepsilon^3\,[x(\ln\Phi)']^3\Bigg]\,,\end{aligned} \tag{21}$$

where $\varepsilon \equiv x^2-x_{\rm A}^2$ and $\alpha \equiv 1-x_{\rm A}^2$. In negative discriminant regions, we denote three solutions of $N=0$ as $M^2=M_{N1}^2$, $M^2=M_{N2}^2$, and $M^2=M_{N3}^2$, for which we assume $M_{N1}^2 \le M_{N2}^2 \le M_{N3}^2$.

Just outside the Alfvén point (i.e., $\varepsilon \ll 1$), since the discriminant $\mathcal{D}_N$ becomes

$$\mathcal{D}_N = -\left(\frac{2x^2}{3x_{\rm A}^4}\alpha\varepsilon\right)^3 + O(\varepsilon^4)\,, \tag{22}$$

the value is always negative, independent of the magnetic field geometry $\Phi(x;\psi)$. Between the Alfvén point and the light cylinder, the curve of $M^2=M_{N1}^2$ is always in the sub-Alfvénic region, and becomes negative outside of the light cylinder, while the curves of $M^2=M_{N2}^2$ and $M^2=M_{N3}^2$ are allowed in the super-Alfvénic region. We thus only consider two branches of $M^2=M_{N2}^2$ and $M^2=M_{N3}^2$.

3.1. *Radial Configuration*

Beyond the Alfvén point, the sign of the discriminant depends on the magnetic field geometry, which is included in higher order terms of equation (22). For a split-monopole field configuration, which leads to $(\ln\Phi)'=0$ and keeps the discriminant $\mathcal{D}_N$ negative, the trans-fast MHD wind has been discussed in Takahashi (1991). We can obtain three branches of solutions; that is, from the condition $D_{\rm FM}=P_1=0$ we have $M_{\rm F}^2=M_{N2}^2(x_{\rm F})$ and from the condition $D_{\rm FM}=P_2=0$ we have $M_{\rm F}^2=M_{N3}^2(x_{\rm F})$ and $M_{\rm F}^2=M_{N1}^2(x_{\rm F})$. The solutions for $N=0$ are plotted by the thick curves in figure 1. For the radial field configuration of $(\ln\Phi)'=0$, the line of $M^2=M_D^2$, which is the solution of $D_{\rm FM}=0$, can intersect with the branch of $M^2=M_{N3}^2$ at either one or two points, or can intersect with the branch of $M^2=M_{N2}^2$ at one point. Each intersection of the curves of $M^2=M_D^2$ and $M^2=M_{N2}^2$ makes a fast magnetosonic point $(x_{\rm F},u_{\rm F})$. If the $D_{\rm FM}=0$ curve does not intersect both branches of $M^2=M_{N3}^2$ and $M^2=M_{N2}^2$ outside of the Alfvén point, no intersection of $N=0$ and $D_{\rm FM}=0$ exists at a finite distance; there is an intersection only at infinity; the Michel's wind solution is included in this category.

The trans-fast MHD wind satisfying the condition $D_{\rm FM}=P_2=0$ at the fast magnetosonic point is ejected from a plasma source located very close to the Alfvén point (see Takahashi 1991). Of course, this wind solution would be rejected as a pulsar wind model because in the pulsar magnetosphere the plasma source would be located near to the star surface ($x_{\rm i} \ll 1$) and the Alfvén point is located very close to the light cylinder ($x_{\rm A} \sim x_{\rm L}=1$). For a rapidly rotating *disk-wind* system, however, such a trans-fast MHD wind might be realized. In this case, since the plasma injected from the disk surface initially rotates with nearly the speed of light, the kinetic part of the wind energy dominates rather than the electromagnetic part of energy. The injected plasma as a disk-wind blows away to a far distant region after passing both the Alfvén point and the fast magnetosonic point. The rotational energy of the plasma converts to the kinetic energy of its poloidal motion; the magnetization parameter σ is initially very small and increases with increasing distance. On the other hand, the trans-fast MHD wind satisfying $D_{\rm FM}=P_1=0$ streams from the star surface with a *constant* velocity throughout the magnetosphere. This wind solution is also not acceptable as a pulsar wind model.

3.2. *Non-Radial Configurations*

We now consider non-radial configurations of the magnetic field lines. The contribution from the second term in the square brackets of equation (12) on the critical conditions becomes very important. If $(\ln\Phi)'$ is negative in the region $x_{\rm A} < x < \infty$, the discriminant $\mathcal{D}_N$

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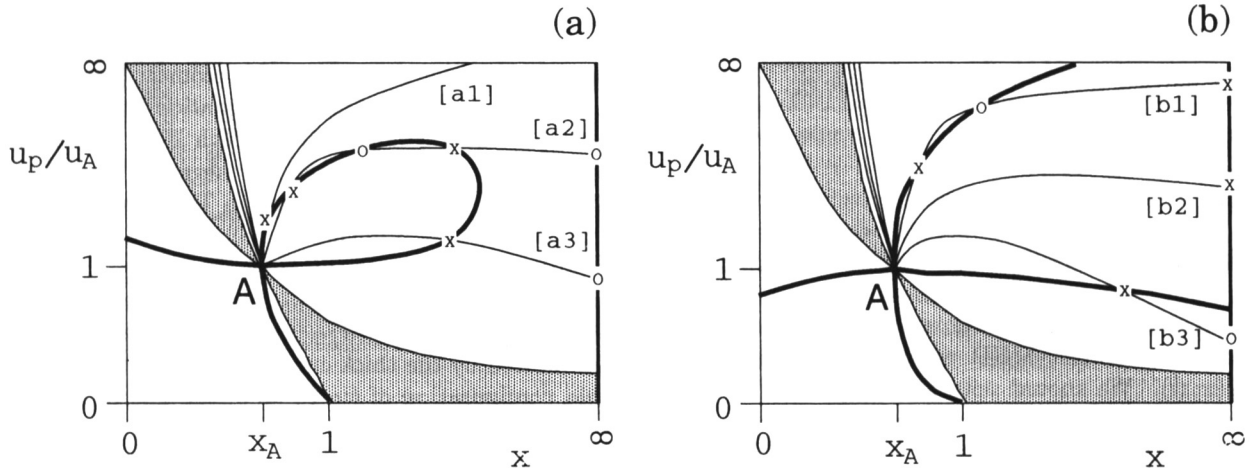


Fig. 2. Behavior of the $D_{FM} = 0$ curve (thin curves) with the $N = 0$ curve (thick curve). The location of the fast magnetosonic point depends on η and the magnetic field configuration: (a) $(\ln\Phi)' < 0$ or (b) $(\ln\Phi)' > 0$. For larger η, the $D_{FM} = 0$ curve labeled '[a1]' and '[a2]' in (a) and labeled '[b1]' in (b) cross the $M_F^2 = M_{N3}^2$ curve, while for smaller η, the $D_{FM} = 0$ curve labeled '[a3]' in (a) and '[b3]' in (b) cross the $M_F^2 = M_{N2}^2$ curve. For the $D_{FM} = 0$ curve labeled '[b2]', there is no intersection of the $D_{FM} = 0$ and $N = 0$ curves at a finite distance; an intersection exists at infinity. The symbols 'X' and 'O' denote X-type (physical) and O-type (unphysical) critical points, respectively.

for $N = 0$ becomes positive at an asymptotic region, while its sign is negative just outside of the Alfvén point. We thus find that the two branches of $M^2 = M_{N2}^2$ and $M^2 = M_{N3}^2$ connect with each other somewhere between $x_A < x < \infty$ and make a $N(x, u_p) = 0$ loop, which starts from and terminates at the Alfvén point (x_A, u_A) on the (x, u_p)-diagram (see figure 1: broken curves). Similarly, outside the Alfvén point, the equation $D_{FM} = 0$ has a solution $u_p = u_{FM}$, where u_{FM} is the fast magnetosonic wave speed (e.g., Takahashi 1991). Because the inclination of $M^2 = M_D^2 \sim O(\varepsilon^{-1/3})$ is greater than that of $M^2 = M_{N2}^2 \sim O(\varepsilon^0)$ and is smaller than that of $M^2 = M_{N3}^2 \sim O(\varepsilon^{-1/2})$, independent of the magnetic field geometry, the curves $N = 0$ and $D_{FM} = 0$ *must* intersect at somewhere located outside the Alfvén point on the (x, u_p)-diagram. That is, a trans-fast MHD wind solution is capable after passing the Alfvén point.

In figures 2a and 2b, we show $D_{FM} = 0$ curves with the $N = 0$ curve for $(\ln\Phi)' < 0$ and $(\ln\Phi)' > 0$, respectively. The shape of the $D_{FM} = 0$ curve depends on η and Φ, while the topological shape of the $N = 0$ line depends on Φ, as can be seen in figure 1; that is, the number of intersections and these locations depend on η and Φ. In figure 2a, for the field configuration of $(\ln\Phi)' < 0$ we can find one or three intersections on the (x, u_p)-plane. The curve marked '[a2]', which gives three intersections, has a larger η than dose curve [a3], and has a smaller η than dose curve [a1], where curves [a1] and [a3] give one intersection. We return this point later in the next subsection.

The appearance of the crossing between the curves $N = 0$ and $D_{FM} = 0$ takes place for any open magnetic field geometry satisfying $(\ln\Phi)'_S < 0$ at the distant nebular shock front ($x_S \gg 1$), even if $(\ln\Phi)' > 0$ is realized somewhere between the plasma source and the nebular shock front. On the other hand, in the case of $(\ln\Phi)'_S \geq 0$, since the discriminant $\mathcal{D}_N$ has a negative value, we may see no loop for $N = 0$ on the (x, u_p)-plane. However, even if the configuration of $(\ln\Phi)'_S > 0$ is held, when $\mathcal{D}_N > 0$ is realized in a region located outside the Alfvén point and inside the nebular shock front, which is caused by $(\ln\Phi)' < 0$, a radius x achieved at $\mathcal{D}_N = 0$ exists; that is, the $N = 0$ loop appears and a trans-fast MHD wind solution is realized. [For a limit of $x_S \to \infty$, we should not overlook that the MHD wind is only realized for $(\ln\Phi)'_\infty \leq 0$ (Chiueh et al. 1991).] The point we wish to emphasize is that the condition $(\ln\Phi)' < 0$ is a necessary condition for creating the $N = 0$ loop. Though we cannot specify a suitable configuration of the magnetic field lines, if we solve the force-balance equation numerically to obtain a magnetic field structure in the magnetosphere self-consistently, we will find that the condition $(\ln\Phi)' < 0$ is automatically satisfied, at least outside of the fast magnetosonic point for a stream function ψ generating the trans-fast MHD wind.

In contrast, for $(\ln\Phi)' > 0$ in the whole range of $x_A < x < \infty$, there are no solutions for $\mathcal{D}_N = 0$. Then, two branches of acceptable physical solutions, $M^2 = M_{N3}^2$ and $M^2 = M_{N2}^2$, do not connect (see figure 1, dotted curves). For the restricted parameters set satisfying the condition $D_{FM} = N = 0$ (see figure 2b, curves [b1] and [b3]), we can find a trans-fast MHD wind solution. However, the wind solution by curve [b1] also represents a disk-wind injected from very close to the Alfvén point,

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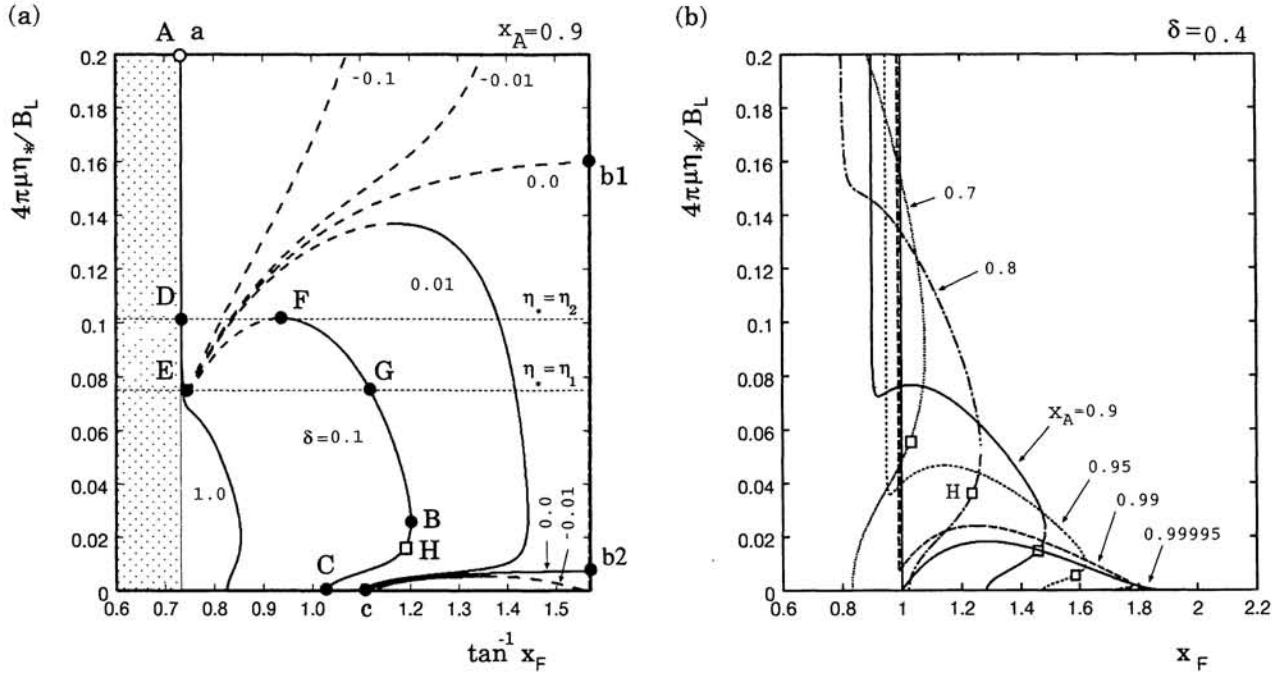


Fig. 3. (a) Particle flux through a flux tube $\eta = \eta_*(x_F; \delta)$ of the trans-fast MHD flow for $x_A = 0.9$. The solid curves give the X-type fast magnetosonic point, while the broken curves give the O-type fast magnetosonic point. For $\eta_1 < \eta_* < \eta_2$ on $\delta = 0.1$ curves, there are two branches of the X-type point: **D**–**E** and **F**–**G**. The points indicated by '**A**' and '**a**' correspond to $x_F \to x_A$ which gives $\eta \to \infty$, while the points indicated by '**C**' and '**c**' denote $\eta_* = 0$. The point indicated by '**B**' on $\delta = 0.1$ curves gives the outer limit of x_F. The points indicated by '**b1**' and '**b2**' correspond to $x_F \to \infty$. (b) The particle flux through a flux tube $\eta = \eta_*(x_F; x_A)$ for $\delta = 0.4$.

and the wind solution by curve [b3] represents a decelerated outflow injected from the star surface with a high velocity (see subsection 4.2).

3.3. *Acceptable Parameter Ranges for Trans-Fast MHD Winds*

Here, to clarify the contributions of the magnetic field geometry on the critical condition at the fast magnetosonic point, we consider a simple model of the poloidal magnetic field from the plasma source to the nebular shock,

$$B_p[x, z(x)] = B_L(\psi) x^{-[2+\delta(\psi)]} , \tag{23}$$

where $\delta(\psi)$ is constant along the field lines and $B_L(\psi)$ is the poloidal magnetic field at the light cylinder. In figures 3 and 4, from equations (19) and (18) with the solutions M^2_{N2} and M^2_{N3}, we plot the particle flux through a flux tube η_* and the total energy E_* of the trans-fast MHD wind, respectively, as a function of x_F for given values of x_A and δ. Rather, we should say that when either η_* or E_* is given by the boundary conditions at the wind origin and the magnetic field configuration is known, the location of the fast magnetosonic point is specified if such a point is possible, while the values of L/E and Ω_F determine the location of the Alfvén point and the light cylinder, respectively. Figures 3a and 4a show the δ-dependence on η_* and E_*, respectively, where these figures are plotted for $x_A = 0.9$ with six configurations of the magnetic flux tube, $\delta = -0.001, -0.1, 0.0, 0.01, 0.1, 1.0$; figures 3b and 4b show the x_A-dependence on η_* and E_*, respectively, where $\delta = 0.4$ and $x_A = 0.7,\ 0.8,\ 0.9,\ 0.95,\ 0.99,\ 0.99995$.

In figures 3 and 4, both η_* and E_* have one or two values for a given x_F; for example, the branches of **A**–**B** and **a**–**b1** are plotted by using the solution $M^2 = M^2_{N3}$ of $N = 0$, and curves **B**–**C** and **b2**–**c** are plotted by using $M^2 = M^2_{N2}$. In the case of $(\ln \Phi)' < 0$ (or $\delta > 0$) the fast magnetosonic point is always allowed at a finite distance for a whole value of η (curve **A**–**C**), which occurs with the $N = 0$ loop formation on (x, u_p)-plane. In the case of $(\ln \Phi)' \geq 0$ (or $\delta \leq 0$), however, the restricted value of η_* only allows the appearance of a fast magnetosonic point (the curves labeled by $-0.1, -0.001, 0.0$ in figure 3a). The Michel's solution, which is the case for the radial field configuration, gives $E = E_0$ with $\delta = 0$; in figure 4a we can see that the $E_* = E_0$ line has no crossing with both curves **b2**–**c** and **a**–**b1**; that is, no fast magnetosonic point exists at a finite distance.

We now consider a trans-fast MHD wind injected with a finite velocity from the plasma source, which is located

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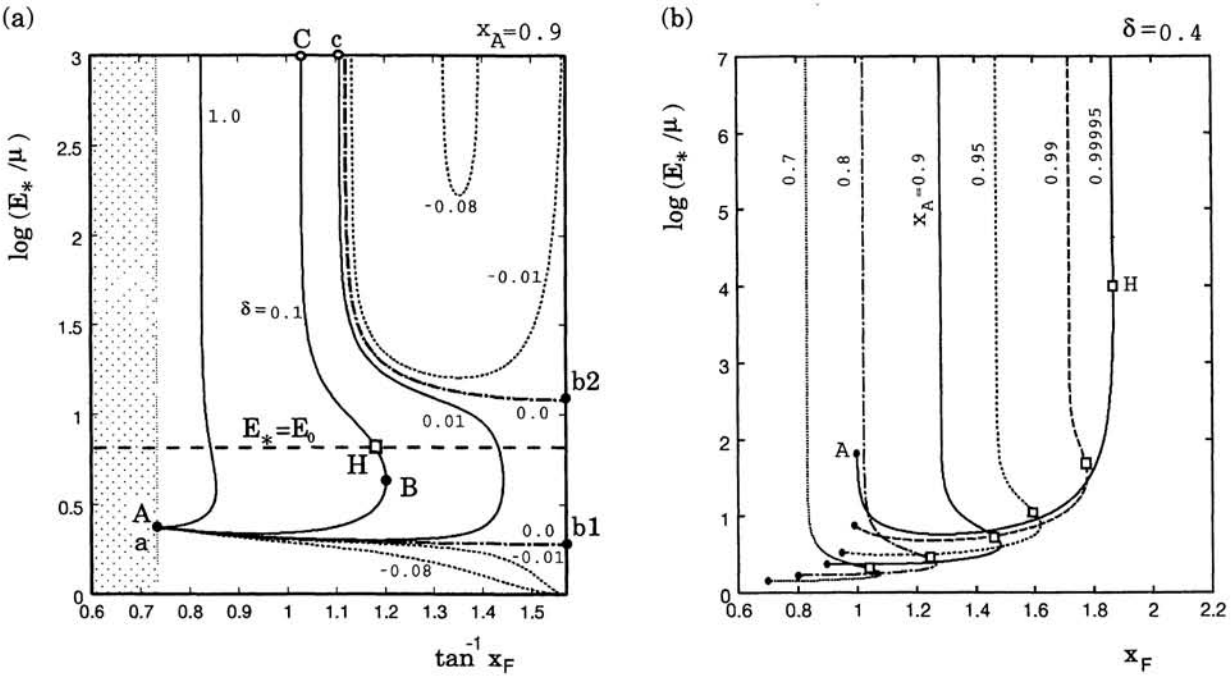


Fig. 4. (a) Total energy $E = E_*(x_F;\delta)$ of the trans-fast MHD flow for $x_A = 0.9$. The solid curves show positive δ, while broken curves show negative δ. The dot-dash-curves show $\delta = 0.0$. Each bold-letter corresponds to that of figure 3. The dotted line denotes $E_* = E_0$; the crossing of this dotted line and the $\delta = 0.1$ curve is indicated by '**H**'. (b) The total energy $E = E_*(x_F;x_A)$ for $\delta = 0.4$.

near to the star surface. To realize such a flow, we must require $E = E_* > E_0$, where $E_0 \equiv \mu/(1-x_A^2)$, which is the energy of the wind injected from the plasma source with zero-velocity; for example, we can see that segments **H**–**C** and **b2**–**c** in figure 4a satisfy this requirement. Thus, from figure 3, we find that there is an upper limit of η_* for the wind (e.g., point **H** for $\delta = 0.1$), which is given by

$$\eta_{\max} = (1-x_A^2)\left[\frac{B_p(M^2-1+x^2)^{1/2}}{4\pi\mu(\mathcal{A}/x)}\right]_{x_F^{\max}}, \qquad (24)$$

where $x_F^{\max}$ denotes the location of the fast magnetosonic point giving $\eta_{\max}$. We also define the fast magnetosonic point $x_F^{\min}$, where $\eta_* = 0$ (e.g., point **C**). The fast magnetosonic point must locate at least between $x_F^{\min}$ and $x_F^{\max}$ for the pulsar wind. Then, figure 3 or figure 4 indicates that the acceptable range of x_F becomes narrow and shifts toward the Alfvén point with the progression of δ, and that it becomes narrow and shifts outward with the progression of x_A.

4. Wind Acceleration and Energy Transport

In this section we discuss the trans-fast MHD pulsar wind from the view of understanding the wind acceleration and energy transport, and lead to the result that the field line of $(\ln\Phi)' \geq 0$ does not give any accelerated trans-fast MHD pulsar wind solutions.

4.1. Near the Injection Region

Let's consider a pulsar wind which is injected from the plasma source ($x_i \ll 1$) with a moderate Lorentz factor. Because of the strong magnetic field and the existence of the upper limit of η, we may say that $\mathcal{M}_i^2 \equiv (M^2/x^2)_i = 4\pi\mu\eta u_i/\Phi_i \ll 1$ and $E_*/E_0 \gg 1$, which are reasonable for typical pulsar parameters ($\mathcal{M}_i^2 \sim 10^{-7}$ for the Crab pulsar; see subsection 4.5). Then, from the poloidal equation, the poloidal velocity far inside the Alfvén point ($x_i \leq x \ll x_A$) is given by

$$\log u_p \sim \log u_i - \frac{1}{2}\log(1-x^2)\,, \qquad (25)$$

where the initial poloidal velocity u_i is defined by

$$u_i^2 = \left(\frac{E_*}{E_0}\right)^2 - 1, \qquad (26)$$

which is denoted by x_A and x_F under a given Φ. Thus, the initial acceleration of this wind is very weak. With increasing distance, this weak acceleration stage would be ended and turn to an acceleration stage.

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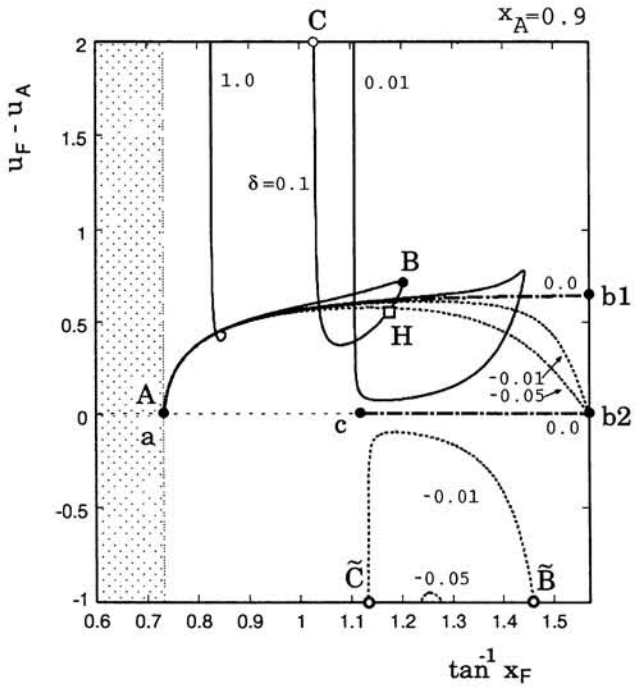


Fig. 5. Wind acceleration around the fast magnetosonic point. The variation in the poloidal velocity $u_F - u_A$ between the Alfvén point ($x_A = 0.9$) and the fast magnetosonic point is plotted. For $(\ln\Phi)' < 0$ ($\delta = 0.01, 0.1, 1.0$) with $\eta_* \ll 1$ (e.g., point **C** for $\delta = 0.1$ curve), a huge accelerated wind is realized, while a decelerated wind is obtained for $(\ln\Phi)' > 0$ ($\delta = -0.01, -0.05$).

4.2. Around the Fast Magnetosonic Point

In figure 5, to discuss the acceleration stage, we plot the differences in the poloidal velocity between the Alfvén point and the fast magnetosonic point $u_F - u_A$ for six values of δ. From the requirement $E_* > E_0$ for the pulsar wind ejected from the pulsar and the fact that the accelerated wind is only obtained for $(\ln\Phi)' < 0$, we pay attention to segment **H**–**C** ($\delta = 0.1$) in figure 5. Referring to figures 3 and 4 for the sequence of solutions, **H**–**C**, we can see that the acceleration between the Alfvén and fast magnetosonic points increases with increasing E and decreasing η. This is due to the corresponding boundary conditions; if the particle flux is reduced, the energy is shared by smaller numbers of particles, so that the kinetic energy of each particle increases; the acceleration increases, of course, with increased input-flux ηE. In connection with subsection 3.3, it must be noted that, to realize a highly accelerated wind, there is no need for $(\ln\Phi)' < 0$ in the whole range of $x_i < x < \infty$; however, there must be a region where $(\ln\Phi)' < 0$, at least around the fast magnetosonic point, to cause $u_F > u_A$.

On the other hand, for $(\ln\Phi)' = 0$ with $E_* > E_0$, we obtain $u_F - u_A = 0$ (segment **c**–**b2**). For such a trans-fast MHD wind, no wind acceleration arises, as mentioned in the previous section. (Note that the fast magnetosonic wave speed turns to decrease with radius between the Alfvén point and the fast magnetosonic point, while the flow velocity is constant.) Further, for $(\ln\Phi)' > 0$ with $E_* > E_0$ (e.g., segment $\tilde{\mathbf{B}}$–$\tilde{\mathbf{C}}$), the wind velocity decreases after passing the Alfvén point. Since the segments **A**–**H** (for $\delta = 0.1$) and **a**–**b1** (for $\delta = 0.0$) denote a solution originating from $0 < x < x_A$ with zero-velocity, it would be available for a disk-wind.

4.3. Asymptotic Region

At the asymptotic region ($x \sim x_S \gg 1$), the total energy of the MHD wind under $(\ln\Phi)'_S \leq 0$ is given by

$$\left(\frac{E_S}{\mu}\right)^2 = (1+u_S^2)\left(\frac{1+\mathcal{M}_S^2}{\mathcal{M}_S^2}\right)^2 + O(x_S^{-2}), \tag{27}$$

where $\mathcal{M}_S^2 \equiv (M^2/x^2)_S = 4\pi\mu\eta u_S/\Phi_S$. Because $E = E_* = E_S$, we can obtain the terminal velocity u_S under a given Φ. We define the acceleration rate ξ of a pulsar wind, where $u_S \gg u_i \gg 1$,

$$\xi \equiv \frac{u_S}{u_i} = \left(\frac{E_0}{\mu}\right)\left(\frac{\mathcal{M}_S^2}{1+\mathcal{M}_S^2}\right) + O(x_S^{-2}), \tag{28}$$

If the poloidal magnetic flux is asymptotically weaken by $\Phi_S \sim o(x^0)$ (or $\delta > 0$), $\mathcal{M}_S^2 \gg 1$ is realized ($\mathcal{M}_S^2 \sim 10^2$ for the Crab pulsar; see subsection 4.5), although both η_* and u_S are restricted by η_{max} and E_*/μ, respectively. From equation (27), we have $u_S \sim (E_S/\mu)$ ($\sim 10^6$ for the Crab pulsar) for the super-fast MHD wind. This means that the total energy of the pulsar wind is dominated by the plasma's kinetic energy at the asymptotic region, while at the injection region the kinetic part of the wind's energy is only a factor of μ/E_0 ($\sim 10^{-4}$ for the Crab pulsar). Thus, the acceleration rate ξ is only determined by the location of the Alfvén point (or L/E); when a highly-accelerated wind where $\xi \gg 1$ is observed, it means that the Alfvén point locates very close to the light cylinder.

Even if $\Phi_S \sim O(x^0)$, the kinetic energy-dominated wind might be realized for $\mathcal{M}_S^2 \gg 1$. However, for $\mathcal{M}_S^2 \ll 1$, the electro-magnetically dominated wind of $u_S \ll (E_S/\mu)$ is obtained. For $\mathcal{M}_S^2 \sim O(x^0)$, the asymptotic velocity u_S is obtained from the cubic equation (27) for given E_S, η, and Φ_S. This relation in an $x_S \to \infty$ limit is plotted in figure 6. The bold-broken curve denotes the Michel's 'minimum energy' solution $E = E_{min}^{\infty} = \mu(1+u_\infty^2)^{3/2}$, which reaches the fast magnetosonic point at infinity. The region marked by 'Super', where a flow has a higher velocity than the Michel's minimum energy solution, is allowed for *super-fast* MHD flows. On the other hand, the left-side region of the bold-broken curve, which is marked by 'Sub', gives asymptotic poloidal velocity of sub-fast MHD flows. To obtain a

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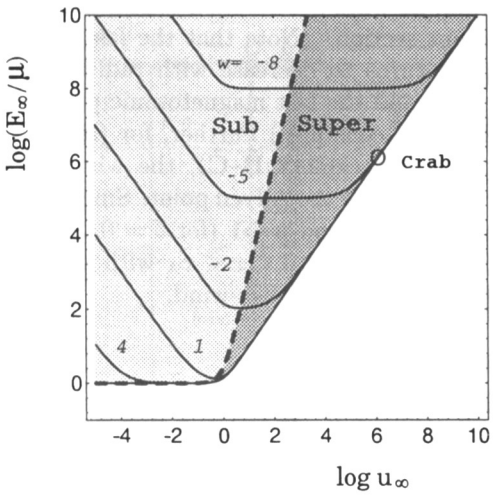


Fig. 6. Asymptotic poloidal velocity u_∞ vs. the total energy E_∞ for several w-values, where $w \equiv \log(4\pi\mu\eta/\Phi_\infty)$ [$\sim (-4)$–(-5) for Crab pulsar wind]. The region indicated by '**Super**' is allowed for super-fast MHD flows, while the region indicated by '**Sub**' is allowed for sub-fast MHD flows. The bold-broken curve denotes the Michel's 'minimum energy' solution. The point labeled by '**Crab**' corresponds to the Crab pulsar wind.

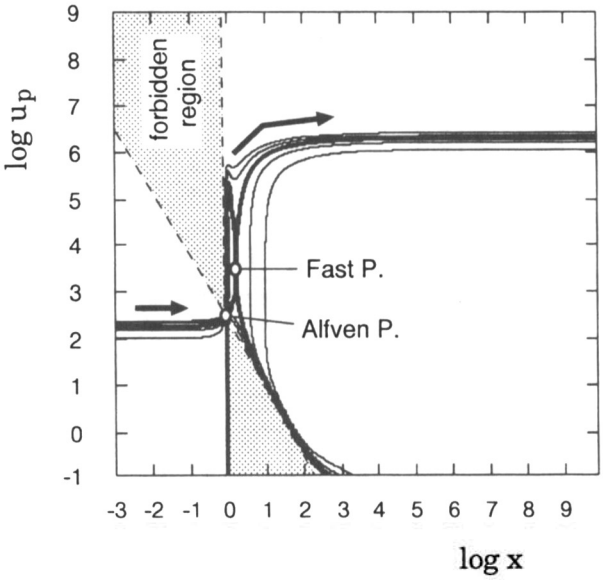


Fig. 7. Pulsar wind solution with $\delta = 0.4$, $x_A = 0.99995$, $x_F = 1.8656662$, where the pulsar surface and the nebular shock front locate $x_i \sim 10^{-2}$ and $x_S \sim 10^9$, respectively. The thick curve with arrows denotes a solution of the trans-fast MHD wind.

highly-accelerated wind (the region marked by 'Super'), the MHD wind must pass through the X-type critical point (that is, the fast magnetosonic point) between the Alfvén point and infinity. As shown the previous section, such situations are available for the curved geometry $(\ln\Phi)' < 0$. When the trans-fast MHD wind is realized, it has energy $E_S \sim E_\infty > E^\infty_{\rm min}$. In a radial field geometry, there is no fast magnetosonic point at a finite distance for the pulsar wind, and only a sub-fast MHD wind is possible; the Michel's minimum energy solution determines the upper-limit of the terminal velocity, $u_\infty = (4\pi\mu\eta/\Phi)^{-1/3}_\infty$.

4.4. *Energy Transport in the Wind*

Next, we consider the energy transport from the electro-magnetic part of the energy to the plasma's kinetic energy in the wind region. The magnetization parameter σ, which is defined as the ratio between the Poynting flux and the total particle energy flux, is given by (e.g., see Camenzind 1986b)

$$\sigma \equiv \frac{-xB_T B_p}{4\pi\mu n u_p u_t} = \frac{M^2-1+x^2}{M^2-1+x_A^2}\mathcal{A} = \frac{x^2-x_A^2}{M^2-1+x_A^2}, \qquad (29)$$

where the toroidal magnetic field B_T is given by $xB_T \equiv \Omega_F B_\phi = -4\pi\eta E\mathcal{A} = -4\pi\eta E[\sigma/(1+\sigma)]$.

The behavior of σ at the asymptotic region have been investigated by Begelman and Li (1994). At the asymptotic region, this ratio becomes $\sigma_S = \Phi_S/(4\pi\mu\eta u_S)$, and for the kinetic energy-dominated wind ($u_S = E/\mu$) it becomes

$$\sigma_S = \Phi_S/(4\pi\eta E)\ . \qquad (30)$$

To realize $\sigma_S \ll 1$, either $\eta \gg 1$ or $\Phi_S \ll 1$ should be required. However, because of the presence of $\eta_{\rm max}$, we must require $\Phi_S \ll 1$ at the asymptotic region. In fact, for $\Phi_S \sim o(x^0)$, we have $\mathcal{M}_S \gg 1$ and the kinetic energy-dominated trans-fast MHD wind is available.

In contrast to this, for any given field configurations satisfying $\Phi_S \sim O(x^0)$, we can also find a highly-accelerated trans-fast MHD wind solution. Such a solution shows that the magnetic part of the energy transports to the kinetic energy part of the energy and an equipartition of energy is achieved; however, the kinetic energy-dominated wind solution cannot be obtained at the nebular shock front.

4.5. *Crab Pulsar Wind*

When a magnetic field line is given, to specify the trans-fast MHD wind solution we must set the four constants of motion (E, L, Ω_F, η), which satisfy the boundary conditions at the plasma source and the nebular shock. However, instead of these four-conserved quantities, we can give the three special radius for the trans-fast MHD wind: x_A, $x_L(=1)$, and x_F; then, by using the poloidal equation we plot a contour map of equi-energy curves on the (x, u_p)-plane, as shown in figure 7, which

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denotes the wind solutions. If the X-type crossing curves appear outside of the Alfvén point, one of the branches indicates the trans-fast MHD wind solution.

We now demonstrate the Crab pulsar wind with the magnetic field assumed by equation (23) to match both the pair-creation model and the nebular shock model. We determine the wind-parameters based on the following reasons:

1. The angular frequency Ω_F determines the location of the light cylinder, $R_L = 1/\Omega_F$, and we have normalized the radius x by R_L. The location of the injection point and the nebular shock are estimated as $x_i \sim 10^{-2}$ and $x_S \sim 10^9$, respectively.

2. From the boundary conditions at the plasma source and the nebular shock, we expect $u_i \sim 10^2$, $u_S \sim 10^6$, and $\xi \approx (1 - x_A^2)^{-1} \sim 10^4$. Thus, the Alfvén radius locates very close to the light cylinder. We set $x_A = 0.9999500$.

3. To realize the trans-fast MHD wind, we require $\delta > 0$. Furthermore, to match the value of σ with 10^{-3}–10^{-4} at the nebular shock, we require that the magnetic flux is weaken at the asymptotic region in accordance with equation (30).

4. Because of the very large total energy, $E = E_* \gg 1$, the location of the fast magnetosonic point should be set very close to x_F^{min}, which is the inner boundary of the acceptable range, $x_F^{min} < x_F < x_F^{max}$. This tendency is clearly seen from the segment **B**–**C** in figure 4. The requirement $x_F \sim x_F^{min}$ also leads a very small η_*/η_{max}.

Then, under $x_A = 0.9999500$, we search for a parameter set (x_F, δ) iteratively to match the boundary conditions of u_i (or σ_i) and σ_S. As a result, we set $x_F = 1.8656662$ and $\delta = 0.4$ and plot figures 7 and 8; thus, we obtain $E_*/\mu \approx 1.71 \times 10^6$ and $4\pi\mu\eta_*/B_L \approx 3.06 \times 10^{-8}$, which are consistent with the results from both the pair-creation model (we assume the multiplicity of pairs is $\sim 10^3$) and the nebular shock model.

In figure 7 we can see that the wind starts from $x_i (\ll 1)$ with $(\ln u_p)_i \sim 2$, which is expected from equation (26). The rapid acceleration of the flow starts from a nearby Alfvén point and occurs within several thousand times of x_F; we have $u_A \approx 2.59 \times 10^2$ and $u_F \approx 2.57 \times 10^3$; then, far beyond the fast magnetosonic point ($x \gg 1$), the wind velocity almost reaches the terminal velocity $(\ln u_p)_S \sim 6$. If we select a slightly larger value of x_F, rather than the above value of x_F, we have a solution in which the initial poloidal velocity is drastically small, rather than that of figure 7. This is because the critical energy E_* is very sensitive to x_F; this tendency can also be clearly seen from the segment **B**–**C** in figure 4.

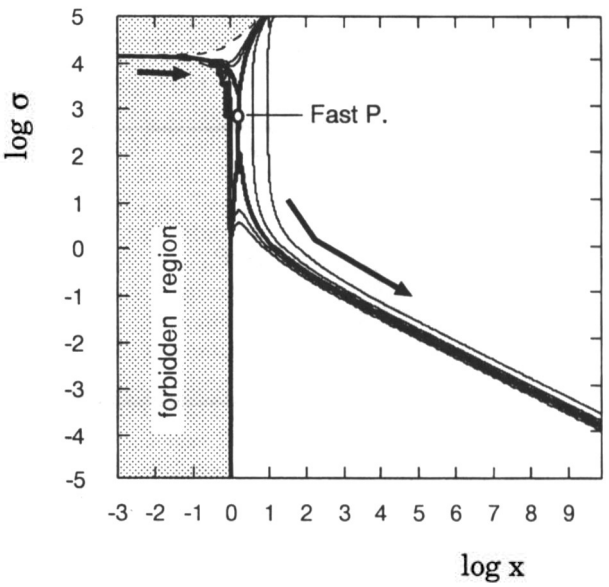


Fig. 8. Energy transport from the electromagnetic part of energy to the kinetic part of energy. The parameters are the same as in figure 7.

Figure 8 shows the behavior of σ for the trans-fast MHD wind. Around the fast magnetosonic point, corresponding to the wind acceleration and the angular momentum gain of the plasma (see figure 9), the initially dominated Poynting flux ($\sigma_i \sim 10^4$) converts to the plasma's kinetic energy. The equipartition of the energy is realized at about 10-times x_F. After realizing the equipartition, it seems that the magnetization parameter decreases as $\sigma \propto x^{-0.4}$ far beyond the fast magnetosonic point. At the nebular shock, we obtain $\sigma_S \sim 10^{-3}$–10^{-4}.

The rapid acceleration is caused by special relativistic effects. Around the fast magnetosonic point located near to the light cylinder, the plasma's inertia forces the magnetic field line to bend drastically toward the counter-rotating direction, so that under the considering poloidal field geometry ($\delta > 0$) a large Lorentz force is generated along the field line. Far beyond the fast magnetosonic point, the toroidal magnetic field decreases as $B_T \propto x^{-(1+\delta)}$ with the progression of distance x, while $B_p \propto x^{-(2+\delta)}$. The field-aligned component of the Lorentz force becomes $(\boldsymbol{j} \times \boldsymbol{B})_\parallel = j_\perp B_T \propto \delta B_T^2$, where $j_\perp$ is the cross-field component of the current. We then see that for $\delta > 0$ the Lorentz force keeps accelerating the wind outward, while for $\delta < 0$ a wind is decelerated (cf., figure 5).

In figure 9, we also plot the fluid parts of energy and angular momentum given by (2) and (3), which are reduced as

$$\frac{\mu u_t}{E} = 1 - \mathcal{A} = \frac{M^2 - 1 + x_A^2}{M^2 - 1 + x^2} \tag{31}$$

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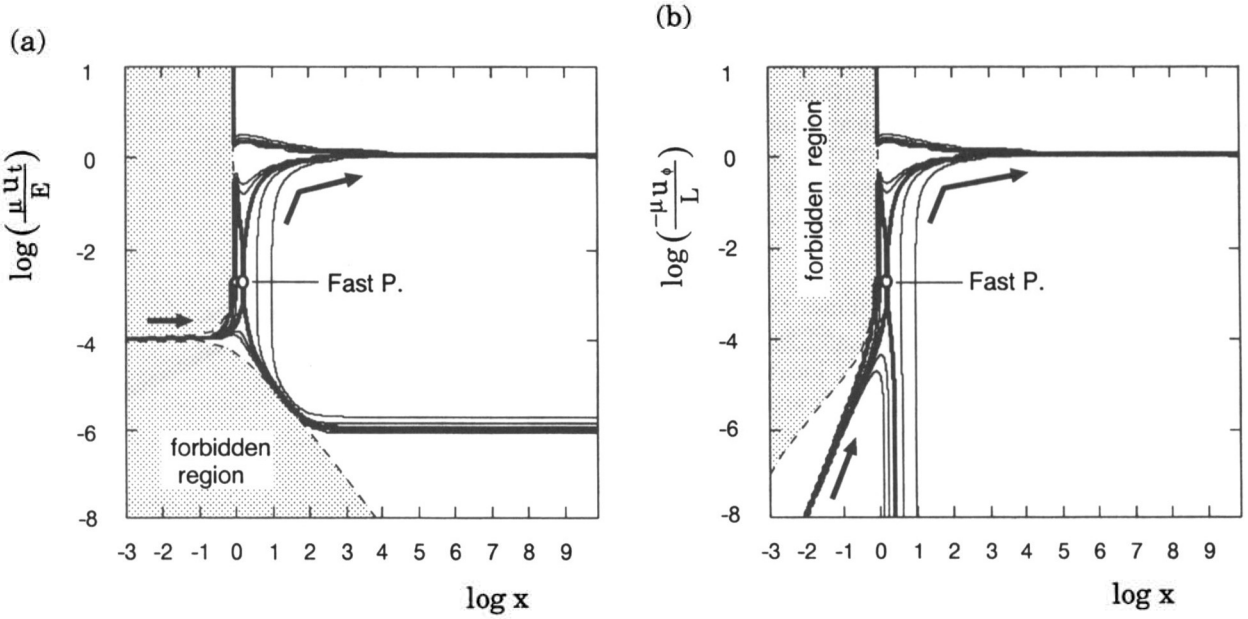


Fig. 9. (a) Energy and (b) angular momentum of the fluid. The parameters are the same as in figure 7.

and

$$\frac{-\mu u_\phi}{L} = 1 - \frac{\mathcal{A}}{x_{\rm A}^2} = \frac{x_{\rm A}^2 M^2 - x^2 M_{\rm A}^2}{x_{\rm A}^2 (M^2 - 1 + x^2)}, \tag{32}$$

respectively. We can see that the initial angular momentum is also magnetically dominated, and that the injected fluid gains angular momentum from the electromagnetic field. The angular momentum transports mainly beyond the fast magnetosonic point, and at almost 10-times $x_{\rm F}$ the angular momentum of the fluid part becomes comparable to that of the electromagnetic field. Then, far beyond the fast magnetosonic point, the angular momentum is dominated by the fluid component, $-\mu u_\phi / L \to 1$, with $\mu u_t / E \to 1$.

5. Concluding Remarks

We have discussed the effect of a magnetic field on the MHD wind structure while considering a simple magnetic field geometry. If a magnetosphere constructs a magnetic field geometry satisfying $(\ln \Phi)' < 0$, a loop is constructed by the lines of $N = 0$ on the (x, u_p)-diagram and the $N = 0$ loop always crosses the line of $D = 0$ somewhere between the Alfvén point and far distant region; that is, the trans-fast MHD wind solution is always capable. We have obtained the restrictions of E_* and η_* on the tarns-fast MHD pulsar wind under fixed wind parameters, δ, Ω_F, L/E; the acceptable ranges of E_* and η_* are expressed in terms of $x_{\rm A}$, $x_{\rm F}$, and δ.

For the pulsar wind, which is highly accelerated and would be the trans-fast MHD wind, the field geometry must satisfy the condition $(\ln \Phi)' < 0$, at least around the fast magnetosonic point. From the viewpoint of the critical condition, there is no restriction for the field geometry over the fast magnetsonic point; at the asymptotic region, however, the field geometry must satisfy $(x^2 B_p)' < 0$ again to satisfy the trans-field equation (Chiueh, Begelman 1991). It seems that the behavior of acceleration and energy-transport in the trans-fast MHD wind do not depends on a detailed function of the field line geometry on the way from the pulsar to the nebular shock, but depends on the divergence of the cross section along the flux tube. However, because the number and locations of the fast magnetosonic point sensitively depend on a function (x-dependence) of the magnetic field geometry, our treatment of the critical conditions would be a very useful tool to solve the magnetosphere with both the poloidal and trans-field equations. While we have only considered monotonically diverging magnetic field lines, the appearance of fast magnetosonic points could be discussed in the same manner along a magnetic field line, even if realistic magnetic field geometries are a more complicated field.

We have also demonstrated a highly-accelerated trans-fast MHD wind. To match the boundary conditions at the wind source which is the outer boundary of the pair-creation region and the MHD shock-front located at the inner boundary of the nebula, we have constructed the magnetic field structure. For the simple field geometry (23), for example, we have evaluated the corresponding value of $\delta = 0.4$. Then, we have demonstrated that the trans-fast MHD wind mainly accelerates between the Alfvén radius and several times of the fast magnetosonic radius, where the kinetic and the electromagnetic energy

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fluxes are comparable; then, on the way to the nebula the electromagnetic energy flux converts to the kinetic energy flux as seen in the Crab pulsar wind.

Thus, for the highly-accelerated trans-fast MHD wind which dominates plasma's kinetic energy flux ($\sigma \gg 1$), we have argued the rapid decline of $x^2 B_p$. However, we should note that matter is required to load on a tiny fraction (10^{-3}–10^{-4}) of all field lines near to the light cylinder. It seems hard to explain the rapid decreasing $x^2 B_p$ for the required decrease in σ. We would need to discuss the magnetic field structure in the vicinity of the pulsar ($\lesssim r_{\rm L}$), which is affected by the plasma production process and remains unclear, and also to treat the equatorial boundary condition carefully as mentioned in Tomimatsu (1994). The asymptotic structure of magnetic fields with the trans-field equation was discussed by Chiueh et al. (1991), Tomimatsu (1994), and Begelman and Li (1994). In contrast to our situation, they argued that the wind acceleration and collimation are logarithmically slow at large distances, and it is very difficult for an ultrarelativistic wind to reach the kinetically dominated asymptotic state at the nebula. Detailed discussions of the global structure of the pulsar magnetosphere based on the observation will be future work.

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References

Arons J. 1979, Space Sci. Rev. 24, 437
Begelman M.C., Li Z.-Y. 1994, ApJ 426, 269
Bekenstein J.D., Oron E. 1978, Phys. Rev. D18, 1809
Beskin V.S., Gurevich A.V., Istomin Ya.N. 1993, Physics of the pulsar magnetosphere (Cambridge University Press, Cambridge) ch4
Camenzind M. 1986a, A&A 156, 137
Camenzind M. 1986b, A&A 162, 32
Cheng K.S., Ho C., Ruderman M. 1986, ApJ 300, 500
Chiueh T., Li Z., Begelman M.C. 1991, ApJ 337, 462
Daugherty J.K., Harding A.K. 1982, ApJ 252, 337
Goldreich P., Julian W.H. 1969, ApJ 157, 869
Gurevich A.V., Istomin Ya.N. 1985, Soviet Phys. JETP 62, 1
Kennel C.F., Coroniti F.V. 1984a, ApJ 283, 694
Kennel C.F., Coroniti F.V. 1984b, ApJ 283, 710
Kennel C.F., Fujimura F.S., Okamoto I. 1983, Geophys. Astrophys. Fluid Dyn. 26, 147
Kennel C.F., Pellat R. 1976, J. Plasma Phys. 15, 335
Kennel C.F., Schmidt G., Wilcox T. 1973, Phys. Rev. Lett. 31, 1364
Kundt W. 1981, in IAU Symposium 95, Pulsars, ed W. Sieber, P. Wielebinski (Reidel, Dordrecht) p57
Kundt W., Krotscheck E. 1980, A&A 83, 1
Max C., Perkins F. 1971, Phys. Rev. Lett. 27, 1342
Mestel L., Shibata S. 1994, MNRAS 271, 621
Michel F.C. 1969, ApJ 157, 1183
Michel F.C. 1982, Rev. Mod. Phys. 54, 1
Rees M.J., Gunn J.E. 1974, MNRAS 167, 1
Ruderman M., Sutherland P.C. 1975, ApJ 196, 51
Shibata S. 1991, ApJ 387, 239
Sturrock P.A. 1971, ApJ 164, 529
Sulkanen M.E., Lovelace R.V.E. 1990, ApJ 350, 732
Tademaru E. 1973, ApJ 183, 625
Takahashi M. 1991, PASJ 43, 569
Takahashi M. 1993, in Plasma Physics and Controlled Nuclear Fusion, ed T.D. Guyenne, J.J. Hunt, ESA SP-351, p329
Tomimatsu A. 1994, PASJ 46, 123

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